

Midterm Assessment Answer Sheet

Semester 1, 2022/2023

Time allowed: 2 hours

Instructions (please read carefully):

- Write down your **Student Number** on the right and using ink or pencil, **shade completely** the corresponding bubbles in the grid for each digit or letter. **DO NOT WRITE YOUR NAME!**
- This answer booklet comprises **ELEVEN (11) pages**, including this cover page.
- This is a **OPEN-SHEET** assessment. You are allowed one A4-sized double-sided cheatsheet.
- Weightage of questions is given in square brackets. The maximum attainable score is 100.
- You are allowed to bring a calculator, but it cannot have any form of external communication capability, i.e. not Wifi- or 4G-enabled. Mobile phones and tablets are not allowed.
- All questions must be answered in the space provided on the answer sheet; no extra sheets will be accepted as answers.
- You are allowed to write with pencils, as long as it is legible.
- Marks may be deducted** for unrecognisable handwriting and/or for not shading the student number properly.
- You must submit only the **ANSWER SHEET** and no other documents. The question set may be used as scratch paper.
- An excerpt of the question may be provided to aid you in answering in the correct box, where applicable. It is not the exact question. You should still refer to the original question in the question booklet.

STUDENT NUMBER														
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For Examiner's Use Only

Question	Marks
Q1	/ 38
Q2	/ 23
Q3	/ 21
Q4	/ 15
Q5	/ 3
Total	/100

Question 1A State Representation

[4 marks]

Question 1B Invariant

[2 marks]

Question 1C Initial and goal states

[4 marks]

Question 1D Which of the following statement(s) is/are true? [4 marks]

- ☐ The search tree is finite.
- ☐ There are possibly many repeated states.
- ☐ There are possibly multiple goal states.
- ☐ We can always find an answer (valid mapping) if we employ the right search algorithm.
- ☐ None of the above.

Question 1E Best search algorithm. [4 marks]

Question 1F To Graph-Search or not? That is the question. [2 marks]

Question 1G Order of Growth (Time) [2 marks]

Question 1H Make it run faster!

[3 marks]

Question 1I Initial candidate solution.

[2 marks]

Question 1J Heuristic function.

[4 marks]

Question 1K Transition functionn.

[4 marks]

Question 1L Which is better: informed search or local search?

[3 marks]

Question 2A Is h_1 admissible?

[2 marks]

Question 2B Is h_2 admissible?

[2 marks]

Question 2C Maximum Branching Factor

[2 marks]

Question 2D h_A admissible and consistent? Explain.

[4 marks]

☐ Admissible ☐ Not admissible

☐ Consistent ☐ Not consistent

Question 2E h_B admissible and consistent? Explain.

[4 marks]

☐ Admissible ☐ Not admissible

☐ Consistent ☐ Not consistent

Question 2F h_C admissible and consistent? Explain.

[4 marks]

☐ Admissible ☐ Not admissible

☐ Consistent ☐ Not consistent

Question 2G Dominant relationships between the heuristics. [5 marks]

Question 3A To save or to spend? [2 marks]

Question 3B Minimal-Depth Tree. [2 marks]

Question 3C What is the entropy?

[2 marks]

Question 3D One-level Tree.

[6 marks]

Question 3E Full Decision Tree.

[5 marks]

Question 3F Min-Sample DT

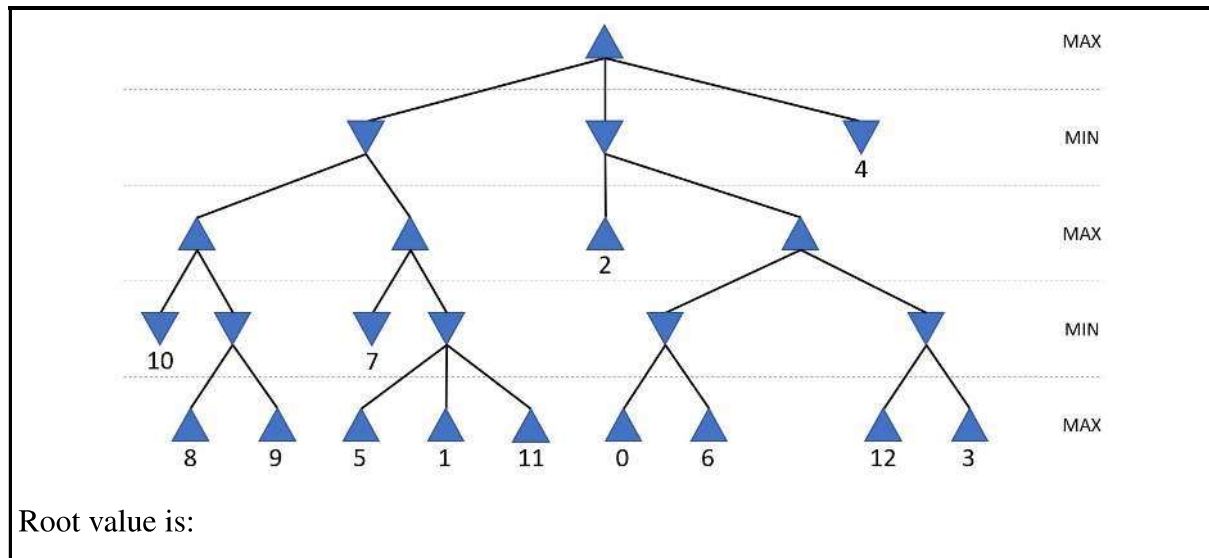
[4 marks]

Question 4A Nim Game Tree.

[6 marks]

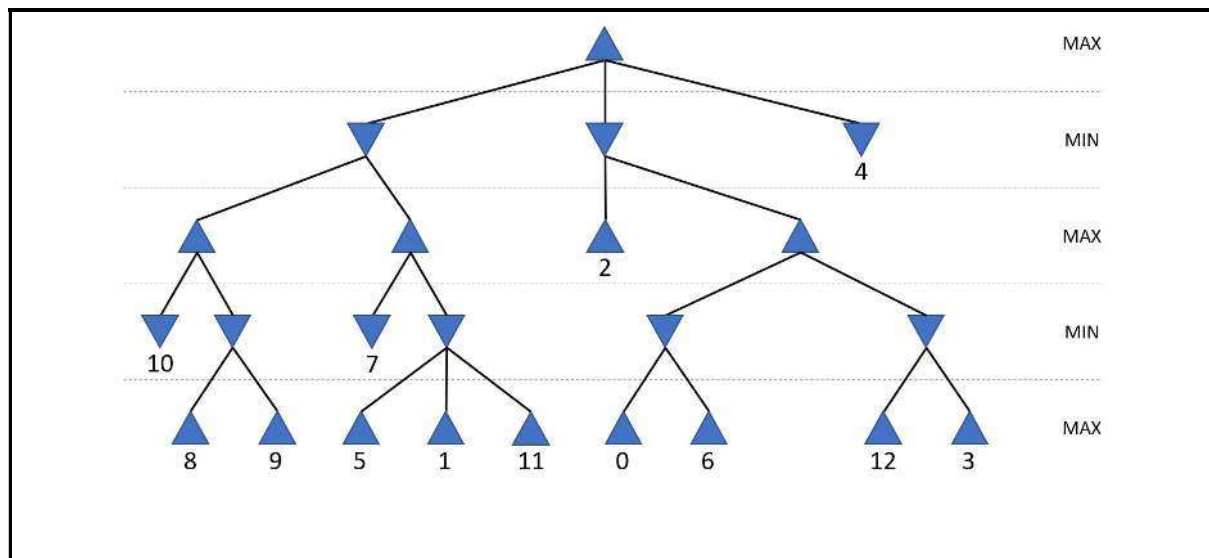
Question 4B Alpha-beta left-to-right.

[5 marks]



Question 4C Alpha-beta right-to-left.

[4 marks]



Question 5 3 most important lessons

[3 marks]

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It may be used as scratch paper.

— END OF ANSWER SHEET —